

Task / Questions and System Output

The questions and the corresponding command/system output below are preserved from the supplied file.

Que 1 →

- Create 1 Public Docker Hub registry named flask_webapp_yourname.

- Clone below repository on your system.

<https://github.com/mmumshad/simple-webapp-docker.git>

- Initialize a local repository & copy the code from above repo to your local repository in

your working branch.

- Once code is copied to the local repository, build the docker image & add meaningful

tags with version 1 and push to Docker Hub registry.

- Once Images are pushed to Docker hub registries, create a directory named kube.

Inside the kube directory create deployment.yaml file with 3 replication , labels app:

flask-webapp , containerPort: 8080 and add the image that you have pushed in Docker

Hub registry.

- Create one service.yaml file with type nodeport & select flask-webapp with port 8080 &

targetPort 8080 with any nodePort between range 30000-32768.

- Once a service is created try accessing the web page in the browser as below. (30011 is

nodeport mentioned in service.yaml). Meanwhile open app.py from your code to understand paths & output.

http://master_ip:30011/

http://master_ip:30011/how are you

- Now , update the app.py from your code and add below route above if `__name__ ==`

`"__main__"` line

```
@app.route('/Who are you')
```

```
def cloudethix():
```

```
    return 'Yes, I am cloudethix, and You !!!'
```

- Once the file is updated , rebuild the docker image & add meaningful tags with version 2

and push to Docker Hub registry.

- Now we have the latest docker image in repo, It's time to roll out a new image. Roll out

the new Image with all three ways one by one.

1. With kubectl set command

2. With kubectl edit deployment

3. With deployment.yaml file modification.

- Run the `# kubectl rollout` command to check status and history.

- Note:- Once above step 1 is done , run `# kubectl rollout undo deployment` command to

rollback the change and then try a second way of rollout.

- In the browser run all three routes & notice the changes.

http://master_ip:30011/

http://master_ip:30011/how are you

http://master_ip:30011/Who are you

- Once done with all above steps , commit all the changes to the remote

repository.

- Capture the snap and prepare a well formatted document.

```
achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ cat Dockerfile app.py
FROM ubuntu:20.04
```

```
RUN apt-get update && apt-get install -y python3 python3-pip
```

```
RUN pip install flask
```

```
COPY app.py /opt/
```

```
ENTRYPOINT FLASK_APP=/opt/app.py flask run --host=0.0.0.0 --port=8080
import os
from flask import Flask
app = Flask(__name__)
```

```
@app.route("/")
def main():
    return "Welcome!"
```

```
@app.route('/how are you')
def hello():
    return 'I am good, how about you?'
```

```
if __name__ == "__main__":
    app.run()
```

```
achal@LAPTOP-HPUOTJF1:kube$ ls
deployment.yml  service.yml
achal@LAPTOP-HPUOTJF1:kube$ cat deployment.yml service.yml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: flask-webapp
  labels:
    app: flask-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: flask-webapp
  template:
    metadata:
      labels:
        app: flask-webapp
    spec:
      containers:
        - name: flask-webapp
          image: docker.io/achalpadol/flask_webapp_achal:v1
          ports:
            - containerPort: 8080
```

```
apiVersion: v1
kind: Service
metadata:
  name: flask-webapp-service
spec:
```

```
type: NodePort
selector:
  app: flask-webapp
ports:
  - port: 8080
    targetPort: 8080
    nodePort: 30011
```

```
achal@LAPTOP-HPUOTJF1:~/Assignment-3$ git clone
https://github.com/mmumshad/simple-webapp-docker.git
```

```
Cloning into 'simple-webapp-docker'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 14 (delta 3), reused 2 (delta 2), pack-reused 7 (from 1)
Receiving objects: 100% (14/14), done.
Resolving deltas: 100% (3/3), done.
achal@LAPTOP-HPUOTJF1:~/Assignment-3$ ls
simple-webapp-docker
```

```
achal@LAPTOP-HPUOTJF1:~/Assignment-3$ cd simple-webapp-docker/
achal@LAPTOP-HPUOTJF1:~/Assignment-3/simple-webapp-docker$ ls
Dockerfile  app.py
```

```
achal@LAPTOP-HPUOTJF1:~/Assignment-3/simple-webapp-docker$ podman build -t
docker.io/achalpadol/flask_webapp_achal:v1 .
STEP 1/5: FROM ubuntu:20.04
Resolved "ubuntu" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/ubuntu:20.04...
Getting image source signatures
Copying blob 13b7e930469f done    |
Copying config b7bab04fd9 done    |
Writing manifest to image destination
STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:2 http://archive.ubuntu.com/ubuntu focal InRelease [265 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages
[1300 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/restricted amd64
Packages [4710 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages
[4368 kB]
Get:6 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:7 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64
Packages [32.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu focal/main amd64 Packages [1275 kB]
```

```
achal@LAPTOP-HPUOTJF1:~/Assignment-3/simple-webapp-docker$ podman images
REPOSITORY                                TAG          IMAGE
ID          CREATED          SIZE
docker.io/achalpadol/flask_webapp_achal    v1
949f9230f443  23 seconds ago  480 MB
```

```
achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ docker push
docker.io/achalpadol/flask_webapp_achal:v1
Getting image source signatures
Copying blob 5a8ac36787a8 done    |
```

```
Copying blob 8168e02bd347 done |
Copying blob 721125e8a881 done |
Copying blob 13b7e930469f skipped: already exists
Copying config 949f9230f4 done |
Writing manifest to image destination
-----

achal@LAPTOP-HPU0TJF1:kube$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
flask-webapp-7c5567647-46v5c        1/1     Running   0           10m
flask-webapp-7c5567647-cm67w        1/1     Running   0           10m
flask-webapp-7c5567647-gp2gg        1/1     Running   0           10m
-----

achal@LAPTOP-HPU0TJF1:kube$ kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)
AGE
flask-webapp-service               NodePort       10.97.197.167    <none>
8080:30011/TCP                     10m
-----

achal@LAPTOP-HPU0TJF1:kube$ curl "http://$(minikube ip):30011/"

Welcome!achal@LAPTOP-HPU0TJF1:kube$ curl "http://$(minikube ip):30011/"

Welcome!achal@LAPTOP-HPU0TJF1:kube$ curl "http://$(minikube ip):30011/Who%20are%20you"e%20you"

<!doctype html>
<html lang=en>
<title>404 Not Found</title>
<h1>Not Found</h1>
<p>The requested URL was not found on the server. If you entered the URL manually please check your spelling and try again.</p>
-----

# updated app.py

achal@LAPTOP-HPU0TJF1:simple-webapp-docker$ cat app.py
import os
from flask import Flask
app = Flask(__name__)

@app.route("/")
def main():
    return "Welcome!"

@app.route('/how are you')
def hello():
    return 'I am good, how about you?'

@app.route('/Who are you')
def cloudethix():
    return 'Yes, I am cloudethix, and You !!!'

if __name__ == "__main__":
    app.run()
-----

achal@LAPTOP-HPU0TJF1:simple-webapp-docker$ podman build -t
docker.io/achalpadol/flask_webapp_achal:v2 .

STEP 1/5: FROM ubuntu:20.04
```

STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip
--> Using cache
eb374c6995cc5c9da545c53818fb15b63b9d3ea3cda6309be11299a130876107
--> eb374c6995cc
STEP 3/5: RUN pip install flask
--> Using cache
f8d2a7f56585a18b5799133c76d7a2f3ed3afe3d4f84a013e73997307bfafd19
--> f8d2a7f56585
STEP 4/5: COPY app.py /opt/
--> ab831a5f60a7
STEP 5/5: ENTRYPOINT FLASK_APP=/opt/app.py flask run --host=0.0.0.0 --
port=8080
COMMIT docker.io/achalpadol/flask_webapp_achal:v2
--> 43ecf906970a
Successfully tagged docker.io/achalpadol/flask_webapp_achal:v2
43ecf906970a397be9d2c4c65836b34b26b8dde28c170c6c15059ee1e5068ebc

achal@LAPTOP-HPU0TJF1:simple-webapp-docker\$ podman images

REPOSITORY	TAG	IMAGE
ID	CREATED	SIZE
docker.io/achalpadol/flask_webapp_achal	v2	
43ecf906970a	4 seconds ago	480 MB

achal@LAPTOP-HPU0TJF1:simple-webapp-docker\$ podman images | grep flask_webapp

docker.io/achalpadol/flask_webapp_achal	v2
43ecf906970a	About a minute ago 480 MB
docker.io/achalpadol/flask_webapp_achal	v1
949f9230f443	14 hours ago 480 MB

achal@LAPTOP-HPU0TJF1:simple-webapp-docker\$ podman push
docker.io/achalpadol/flask_webapp_achal:v2
Getting image source signatures
Copying blob 6ac90cd9bb76 done |
Copying blob 13b7e930469f skipped: already exists
Copying blob d59147826128 skipped: already exists
Copying blob bde202e56e37 skipped: already exists
Copying config 43ecf90697 done |
Writing manifest to image destination

achal@LAPTOP-HPU0TJF1:simple-webapp-docker\$ kubectl get deployment flask-webapp -n cloudeithx-achal -o wide

NAME	READY	UP-TO-DATE	AVAILABLE	AGE	CONTAINERS	IMAGES
flask-webapp	3/3	3	3	43m	flask-webapp	
docker.io/achalpadol/flask_webapp_achal:v1					app=flask-webapp	

achal@LAPTOP-HPU0TJF1:simple-webapp-docker\$ kubectl describe deployment flask-webapp -n cloudeithx-achal | grep Image
Image: docker.io/achalpadol/flask_webapp_achal:v1

achal@LAPTOP-HPU0TJF1:simple-webapp-docker\$ kubectl set image
deployment/flask-webapp \
flask-webapp=docker.io/achalpadol/flask_webapp_achal:v2

deployment.apps/flask-webapp image updated

achal@LAPTOP-HPU0TJF1:simple-webapp-docker\$ kubectl rollout status
deployment/flask-webapp -n cloudeithx-achal

```
deployment "flask-webapp" successfully rolled out
-----
achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl rollout history
deployment/flask-webapp -n cloudethix-achal
deployment.apps/flask-webapp
REVISION  CHANGE-CAUSE
1          <none>
2          <none>
-----

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl describe deployment flask-
webapp -n cloudethix-achal | grep Image
      Image:          docker.io/achalpadol/flask_webapp_achal:v2
-----

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl get deployment flask-
webapp -o wide

NAME                READY   UP-TO-DATE   AVAILABLE   AGE    CONTAINERS      IMAGES
SELECTOR
flask-webapp        3/3     3            3           50m    flask-webapp    docker.io/achalpadol/flask_webapp_achal:v2
app=flask-webapp
-----

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl rollout undo
deployment/flask-webapp

Warning: resource deployments/flask-webapp was previously managed with
'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-
applied-configuration annotation, which may cause unexpected behavior on
future 'kubectl apply' operations. Consider using 'kubectl apply' with your
previous configuration file instead.
deployment.apps/flask-webapp rolled back
-----

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl get deployment flask-
webapp -o wide

NAME                READY   UP-TO-DATE   AVAILABLE   AGE    CONTAINERS      IMAGES
SELECTOR
flask-webapp        3/3     3            3           66m    flask-webapp    docker.io/achalpadol/flask_webapp_achal:v1
app=flask-webapp
-----

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl rollout status
deployment/flask-webapp -n cloudethix-achal
deployment "flask-webapp" successfully rolled out
-----

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl describe deployment flask-
webapp -n cloudethix-achal | grep Image
      Image:          docker.io/achalpadol/flask_webapp_achal:v1
-----

# Step edit file
kubectl edit deployment flask-webapp

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl edit deployment flask-
webapp
deployment.apps/flask-webapp edited

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl rollout status
deployment/flask-webapp
deployment "flask-webapp" successfully rolled out

achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl rollout history
```

```
deployment/flask-webapp
deployment.apps/flask-webapp
```

REVISION	CHANGE-CAUSE
3	<none>
4	<none>

```
# undo again to check third step
```

```
-----
achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl rollout undo
deployment/flask-webapp
```

```
Warning: resource deployments/flask-webapp was previously managed with
'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-
applied-configuration annotation, which may cause unexpected behavior on
future 'kubectl apply' operations. Consider using 'kubectl apply' with your
previous configuration file instead.
deployment.apps/flask-webapp rolled back
```

```
-----
achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl rollout status
deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
```

```
achal@LAPTOP-HPUOTJF1:simple-webapp-docker$ kubectl describe deployment flask-
webapp -n cloudeithx-achal | grep Image
      Image:          docker.io/achalpadol/flask_webapp_achal:v1
```

```
-----
#with deployment.yaml file modification.
```

```
achal@LAPTOP-HPUOTJF1:kube$ vim deployment.yaml
achal@LAPTOP-HPUOTJF1:kube$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: flask-webapp
  labels:
    app: flask-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: flask-webapp
  template:
    metadata:
      labels:
        app: flask-webapp
    spec:
      containers:
        - name: flask-webapp
          image: docker.io/achalpadol/flask_webapp_achal:v2
          ports:
            - containerPort: 8080
```

```
-----
achal@LAPTOP-HPUOTJF1:kube$ kubectl apply -f deployment.yaml
deployment.apps/flask-webapp configured
```

```
achal@LAPTOP-HPUOTJF1:kube$ kubectl rollout status deployment/flask-webapp
deployment "flask-webapp" successfully rolled out
```

```
achal@LAPTOP-HPUOTJF1:kube$ kubectl rollout history deployment/flask-webapp
```

```
deployment.apps/flask-webapp
```

REVISION	CHANGE-CAUSE
----------	--------------

5	<none>
---	--------

6	<none>
---	--------

Que 3 →

- Create 1 Public Docker Hub registry named cloudethix_configmap_yourname.

- Clone below repository on your system.

<https://github.com/zembutsu/docker-sample-nginx.git>

- Initialize a local repository & copy the code from above repo to your local repository in

the working branch.

- Once code is copied , build a docker image from docker file and add meaningful tags

and push to docker hub repository.

- Once Images are pushed to Docker hub registries, create a directory named kube.

Inside the kube directory create deployment.yaml file with 3 replication , labels app:

frontend-webapp , containerPort: 80 and add the image that you have pushed in Docker

Hub registry.

- Create one service.yaml file with type nodeport & select frontend-webapp pod with port

80 & targetPort 80 with any nodePort between range 30000-32768.

- Once the service is created try accessing the web page in the browser as below. Notice

the changes & take the snap.

- Now create a configmap.yaml file with below data & delete the deployment that you have created.

```
<html>
```

```
<body>
```

```
<h1> I am Cloudethix Team, Are you ?!! </h1>
```

```
Version: 1.1
```

```
</body>
```

```
</html>
```

- Then update the same deployment.yaml file and mount configmap as volume on container using volumeMounts with mountPath /usr/share/nginx/html/

- Now it's time to create configmap & deployment. Once created , try to access the

webpage in the browser & confirm that the index page is the same as we have in configmap.

```
achal@LAPTOP-HPUOTJF1:configmap-dir$ ls
```

```
Dockerfile  LICENSE  README.md  default.conf  index.html
```

```
achal@LAPTOP-HPUOTJF1:configmap-dir$ git init
```

hint: Using 'master' as the name for the initial branch. This default branch name

hint: will change to "main" in Git 3.0. To configure the initial branch name

hint: to use in all of your new repositories, which will suppress this warning,

hint: call:

hint:


```
hint:  git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:  git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
```

```
Initialized empty Git repository in /home/achal/configmap-dir/.git/
```

```
-----
achal@LAPTOP-HPU0TJF1:configmap-dir$ podman build -t
docker.io/achalpadol/cloudethix_configmap_achal:v1 .
```

```
WARN[0000] "/" is not a shared mount, this could cause issues or missing
mounts with rootless containers
```

```
STEP 1/3: FROM nginx:alpine
```

```
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
```

```
--> Using cache
```

```
090b497e1161956bc69c29ed05f63003fabe8615f06a8b8a09cc43f4e2783826
```

```
--> 090b497e1161
```

```
STEP 3/3: COPY index.html /usr/share/nginx/html/
```

```
--> Using cache
```

```
97ebac17772705f64c7fb37106bd497011c968ff7fdf48c4a0f6d8233a32ea5b
```

```
COMMIT docker.io/achalpadol/cloudethix_configmap_achal:v1
```

```
--> 97ebac177727
```

```
Successfully tagged docker.io/achalpadol/cloudethix_configmap_achal:v1
```

```
Successfully tagged docker.io/achalpadol/cloudethix-nginx-achal:v1
```

```
Successfully tagged localhost/achalpadol/cloudethix-nginx-achal:v1
```

```
Successfully tagged localhost/cloudethix-nginx-achal:v1
```

```
97ebac17772705f64c7fb37106bd497011c968ff7fdf48c4a0f6d8233a32ea5b
```

```
-----
achal@LAPTOP-HPU0TJF1:configmap-dir$ podman images | grep cloudethix_configmap
docker.io/achalpadol/cloudethix_configmap_achal      v1
97ebac177727  24 hours ago  64.2 MB
```

```
achal@LAPTOP-HPU0TJF1:configmap-dir$ podman push
docker.io/achalpadol/cloudethix_configmap_achal:v1
```

```
Getting image source signatures
```

```
Copying blob 0d62d88506ba skipped: already exists
```

```
Copying blob d94291c26261 skipped: already exists
```

```
Copying blob 1ed8e39a7434 skipped: already exists
```

```
Copying blob 8cdfef4f23778 skipped: already exists
```

```
Copying blob 0ea935727878 skipped: already exists
```

```
Copying blob 55afa1ecc21d skipped: already exists
```

```
Copying blob fb597529c916 skipped: already exists
```

```
Copying blob da4dea1b00af skipped: already exists
```

```
Copying blob 0cc7f3a48f3f skipped: already exists
```

```
Copying blob 4b794759b385 skipped: already exists
```

```
Copying config 97ebac1777 done |
```

```
Writing manifest to image destination
```

```
-----
achal@LAPTOP-HPU0TJF1:kube$ cat deployment.yaml service.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: frontend-webapp
```

```
  namespace: cloudethix-achal
```

```
  labels:
```

```
    app: frontend-webapp
```

```
spec:
  replicas: 3

  selector:
    matchLabels:
      app: frontend-webapp

  template:
    metadata:
      labels:
        app: frontend-webapp

    spec:
      containers:
        - name: frontend-webapp
          image: docker.io/achalpadol/cloudethix_configmap_achal:v1
          ports:
            - containerPort: 80
```

```
apiVersion: v1
kind: Service
metadata:
  name: frontend-webapp-service
  namespace: cloudethix-achal
```

```
spec:
  type: NodePort

  selector:
    app: frontend-webapp

  ports:
    - port: 80
      targetPort: 80
      nodePort: 30011
```

```
achal@LAPTOP-HPU0TJF1:kube$ cat configmap.yaml
```

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: frontend-html
  namespace: cloudethix-achal
```

```
data:
  index.html: |
    <html>
    <body>
    <h1> I am Cloudethix Team, Are you ?!! </h1>
    Version: 1.1
    </body>
    </html>
```

```
# updated deployment.yml
```

```
achal@LAPTOP-HPU0TJF1:kube$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-webapp
```

```

namespace: cloudethix-achal
labels:
  app: frontend-webapp

spec:
  replicas: 3

  selector:
    matchLabels:
      app: frontend-webapp

  template:
    metadata:
      labels:
        app: frontend-webapp

    spec:
      containers:
        - name: frontend-webapp
          image: docker.io/achalpadol/cloudethix_configmap_achal:v1
          ports:
            - containerPort: 80

          volumeMounts:
            - name: nginx-html
              mountPath: /usr/share/nginx/html/

      volumes:
        - name: nginx-html
          configMap:
            name: frontend-html

```

Que 4 →

- Create 1 Public Docker Hub registry named cloudethix_multicontainer_yourname.
- Clone below repository on your system.
<https://github.com/janakiramm/Kubernetes-multi-container-pod.git>
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branches.
- Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.
- Now go to the deploy directory and notice the files.
- Here, web-pod-1.yml file will create the pod with two containers (Multi container). Take a note of labels , name of containers and ports. Also, please make sure you will update the python container image that you have pushed to your docker registry.
- Now, open web-svc.yml file and notice service Type , selectors & targetPort. Apply the file.
- Now open db-pod.yml & notice the labels , name , Image, containerPort and apply the file.
- Now open the db-svc.yml file and notice service Type , selectors & targetPort. Apply the file.
- Once above files are applied , Check that the Pods and Services are created using

command line or lens.

- Now , from the command line run below urls & notice the changes.

```
# curl http://$NODE_IP:$NODE_PORT/init
```

Initialize the database with sample schema

- Now it's time to Insert some sample data. Make sure you will use correct \$NODE_IP:\$NODE_PORT

```
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "1",  
"user": "John Doe"}' http://$NODE_IP:$NODE_PORT/users/add
```

```
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "2",  
"user": "Jane Doe"}' http://$NODE_IP:$NODE_PORT/users/add
```

```
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "3",  
"user": "Bill Colls"}' http://$NODE_IP:$NODE_PORT/users/add
```

```
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "4",  
"user": "Mike Taylor"}' http://$NODE_IP:$NODE_PORT/users/add
```

- Now access the data that we have added to database using below command.

```
# curl http://$NODE_IP:$NODE_PORT/users/1
```

- The second time you access the data, it appends '(c)' indicating that it is pulled from the

Redis cache.

- Also, try to access mysql shell i.e db pod & run select * from the users table. check

app.py for DB related information.

- Prepare proper documentation in brief & write start to end flow. Refer below link if you

face any issues.

<https://github.com/janakiramm/Kubernetes-multi-container-pod>

```
-----  
achal@LAPTOP-HPUOTJF1:kube$ cd
```

```
achal@LAPTOP-HPUOTJF1:~$ git clone https://github.com/janakiramm/Kubernetes-  
multi-container-pod.git
```

Cloning into 'Kubernetes-multi-container-pod'...

remote: Enumerating objects: 51, done.

remote: Total 51 (delta 0), reused 0 (delta 0), pack-reused 51 (from 1)

Receiving objects: 100% (51/51), 88.14 KiB | 9.79 MiB/s, done.

Resolving deltas: 100% (21/21), done.

```
-----  
achal@LAPTOP-HPUOTJF1:~$ cd Kubernetes-multi-container-pod
```

```
achal@LAPTOP-HPUOTJF1:Kubernetes-multi-container-pod$ ls
```

Build Deploy README.md multi-container-pod.png

```
chal@LAPTOP-HPUOTJF1:Kubernetes-multi-container-pod$ cd ~
```

```
achal@LAPTOP-HPUOTJF1:~$ mkdir cloudethix_multicontainer_achal
```

```
achal@LAPTOP-HPUOTJF1:~$ cd cloudethix_multicontainer_achal/
```

```
achal@LAPTOP-HPUOTJF1:cloudethix_multicontainer_achal$ git init
```

hint: Using 'master' as the name for the initial branch. This default branch name

hint: will change to "main" in Git 3.0. To configure the initial branch name

hint: to use in all of your new repositories, which will suppress this warning,

hint: call:

hint:

hint: git config --global init.defaultBranch <name>

hint:

```
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/home/achal/cloudethix_multicontainer_achal/.git/
-----
achal@LAPTOP-HPUOTJF1:cloudethix_multicontainer_achal$ git branch -M master

achal@LAPTOP-HPUOTJF1:cloudethix_multicontainer_achal$ cp -r ~/Kubernetes-
multi-container-pod/* .

achal@LAPTOP-HPUOTJF1:cloudethix_multicontainer_achal$ ls
Build  Deploy  README.md  multi-container-pod.png

achal@LAPTOP-HPUOTJF1:cloudethix_multicontainer_achal$ cd Build/

achal@LAPTOP-HPUOTJF1:Build$ ls

Dockerfile  app.py  docker-compose.yml  requirements.txt
-----
achal@LAPTOP-HPUOTJF1:Build$ podman build -t
docker.io/achalpadol/cloudethix_multicontainer_achal:v1 .
STEP 1/3: FROM python:2.7-onbuild
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:2.7-onbuild...
Getting image source signatures
Copying blob 0c3de61682aa done    |
Copying blob 46dde23c37b3 done    |
Copying blob d660b1f15b9b done    |
Copying blob e7428f935583 done    |
Copying blob 56f10ddf1173 done    |
Copying blob 6ebaeb074589 done    |
Copying blob 4473537c621d done    |
Copying blob 3106f7df3d1c done    |
Copying blob 3de1c6ceef68 done    |
Copying config 3f246dd60a done    |
Writing manifest to image destination
STEP 2/6: COPY requirements.txt /usr/src/app/
--> 88f14b23e568
STEP 3/6: RUN pip install --no-cache-dir -r requirements.txt
Collecting flask (from -r requirements.txt (line 1))
  Downloading https://files.pythonhosted.org/packa
-----
achal@LAPTOP-HPUOTJF1:Build$ podman push
docker.io/achalpadol/cloudethix_multicontainer_achal:v1
Getting image source signatures
Copying blob 6ebaeb074589 skipped: already exists
Copying blob 0c3de61682aa skipped: already exists
Copying blob 46dde23c37b3 skipped: already exists
Copying blob d660b1f15b9b skipped: already exists
Copying blob 56f10ddf1173 skipped: already exists
Copying blob e7428f935583 skipped: already exists
Copying blob 4f0e84c5ce6c done    |
Copying blob 4cf08394a795 done    |
Copying blob 0ba33796ade2 done    |
```

```
Copying blob 3de1c6ceef68 skipped: already exists
Copying blob 3106f7df3d1c skipped: already exists
Copying blob 4473537c621d skipped: already exists
Copying config 7997bb4680 done |
Writing manifest to image destination

-----

achal@LAPTOP-HPUOTJF1:Deploy$ cat web-pod-1.yml
apiVersion: "v1"
kind: Pod
metadata:
  name: web1
  labels:
    name: web
    app: demo
spec:
  containers:
    - name: redis
      image: redis
      ports:
        - containerPort: 6379
          name: redis
          protocol: TCP
    - name: python
      image: docker.io/achalpadol/cloudethix_multicontainer_achal:v1
      env:
        - name: "REDIS_HOST"
          value: "localhost"
      ports:
        - containerPort: 5000
          name: http
          protocol: TCP

-----

achal@LAPTOP-HPUOTJF1:Deploy$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
cloudethix-static-minikube         0/1     Unknown   0           21m
mysql                              1/1     Running   0           13m
web1                                2/2     Running   0           54s

-----

achal@LAPTOP-HPUOTJF1:Deploy$ kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)
AGE
c
mysql                              ClusterIP      10.98.67.16     <none>           3306/TCP
78m
web                                NodePort       10.97.218.252   <none>           80:32234/TCP
16m

-----

achal@LAPTOP-HPUOTJF1:~$ curl http://localhost:8080/init
DB Init doneachal@LAPTOP-HPUOTJF1:~$

-----

achal@LAPTOP-HPUOTJF1:~$ curl http://localhost:8080/init
DB Init doneachal@LAPTOP-curl -i -H "Content-Type: application/json" -X POST -
d '{"uid": "1",{"uid": "1",
"user":"John Doe"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:57:59 GMT
```

```
Addedachal@LAPTOP-HPU0TJFcurl -i -H "Content-Type: application/json" -X POST -
d '{"uid": "2", "2",
"user":"Jane Doe"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:58:46 GMT
```

```
Addedachal@LAPTOP-HPU0TJFcurl -i -H "Content-Type: application/json" -X POST -
d '{"uid": "3", "3",
"user":"Bill Colls"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:59:24 GMT
```

```
Addedachal@LAPTOP-HPU0TJFcurl -i -H "Content-Type: application/json" -X POST -
d '{"uid": "4", "4",
"user":"Mike Taylor"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:59:50 GMT
```

```
Addedachal@LAPTOP-HPU0TJFcurl http://localhost:8080/users/1ers/1
John Doeachal@LAPTOP-HPU0TJF1:~$
```

```
chal@LAPTOP-HPU0TJF1:~$ kubectl exec -it mysql -n cloudehix-achal -- mysql -u
root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 5
Server version: 5.7.25 MySQL Community Server (GPL)
```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> show databases:
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual
that corresponds to your MySQL server version for the right syntax to use near
': ' at line 1
mysql> SHOW DATABASES;
+-----+
| Database          |
+-----+
| information_schema |
| USERDB            |
| mysql             |
| performance_schema |
| sys               |
+-----+
```

5 rows in set (0.01 sec)

mysql> use USERDB;

Reading table information for completion of table and column names

You can turn off this feature to get a quicker startup with -A

Database changed

mysql> show tables;

+-----+

| Tables_in_USERDB |

+-----+

| users |

+-----+

1 row in set (0.00 sec)

mysql> select * from users;

+-----+-----+

| ID | USER |

+-----+-----+

| 1 | John Doe |

| 2 | Jane Doe |

| 3 | Bill Colls |

| 4 | Mike Taylor |

+-----+-----+

4 rows in set (0.00 sec)

Create 1 Public Docker Hub registry named cloudehix_Initcontainer_yourname.

- Clone below repository on your system.

<https://github.com/janakiramm/simpleapp.git>

- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.

- Once code is copied , go to the Build directory and build docker image from docker file

and add meaningful tags and push to docker hub repository.

- Once Images are pushed to Docker hub registries, create a directory named kube.

Inside the kube directory create deployment.yaml file with 3 replication , label app:

simpleapp-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.

- Create one service.yaml file with type nodeport & select simpleapp-webapp pod with port

80 & targetPort 80 with any nodePort between range 30000-32768.

- Open the webpage in the browser and notice the changes and capture the snap.

- Then delete the deployment that you have just created.

- Update the deployment.yaml file and add volumeMounts with mountPath /usr/share/nginx/html from emptyDir: {} volume.

- Once above changes are added, add initContainers block with below parameters. Also

add volumeMounts for Init Container with mountPath "/work-dir" from emptyDir: {}

volume.

initContainers:

- name: install

image: busybox:1.28

command:

- wget

- "-O"

- "/work-dir/index.html"

- http://info.cern.ch

volumeMounts:

- name: workdir

mountPath: "/work-dir"

- Add volumes with emptyDir: {} in deployment.yaml file.

- Once the deployment.yaml file is ready, create the deployment & access the page in the

browser and notice the changes.

- Prepare a well formatted document and write your understanding step by step.

```
achal@LAPTOP-HPUOTJF1:~$ git clone https://github.com/janakiramm/simpleapp.git
Cloning into 'simpleapp'...
```

```
remote: Enumerating objects: 47, done.
```

```
remote: Total 47 (delta 0), reused 0 (delta 0), pack-reused 47 (from 1)
```

```
Receiving objects: 100% (47/47), 8.20 KiB | 599.00 KiB/s, done.
```

```
Resolving deltas: 100% (9/9), done.
```

```
achal@LAPTOP-HPUOTJF1:~$ cd simpleapp/
```

```
achal@LAPTOP-HPUOTJF1:simpleapp$ ls
```

```
Dockerfile  html  wrapper.sh
```

```
achal@LAPTOP-HPUOTJF1:simpleapp$ cd ..
```

```
achal@LAPTOP-HPUOTJF1:~$ mkdir cloudethix_Initcontainer_achal
```

```
achal@LAPTOP-HPUOTJF1:~$ cd cloudethix_Initcontainer_achal/
```

```
achal@LAPTOP-# cp -r ~/simpleapp/* .
```

```
achal@LAPTOP-HPUOTJF1:cloudethix_Initcontainer_achal$ ls
```

```
Dockerfile  html  wrapper.sh
```

```
achal@LAPTOP-# podman images
```

REPOSITORY	TAG	IMAGE
ID	CREATED	SIZE
docker.io/achalpadol/cloudethix_multicontainer_achal	v1	
7997bb46809e	3 hours ago	712 MB

```
achal@LAPTOP-HPUOTJF1:cloudethix_Initcontainer_achal$ podman push
```

```
docker.io/achalpadol/cloudethix_initcontainer_achal:v1
```

```
Getting image source signatures
```

```
Copying blob 6310eb16bf42 skipped: already exists
```

```
Copying blob a44b5c8be616 skipped: already exists
```

```
Copying blob c12f394dea35 skipped: already exists
```

```
Copying blob 07db7bf2649b skipped: already exists
```

```
Copying blob 02fc02c4ab8d skipped: already exists
```

```
Copying blob 956faab5efb3 skipped: already exists
```

```
Copying blob 95f43ddacf80 done |
```

```
Copying blob f530d0743aea done |
```

```
Copying blob f340c1b7c1d6 skipped: already exists
```

```
Copying config 6c9ec7c4cb done |
```

```
Writing manifest to image destination
```

```
achal@LAPTOP-HPUOTJF1:kube$ ls
```

```
deployment.yaml  service.yaml
```

```
achal@LAPTOP-HPUOTJF1:kube$ cat deployment.yaml service.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
  name: simpleapp-webapp
spec:
  replicas: 3
  selector:
    matchLabels:
      app: simpleapp-webapp
  template:
    metadata:
      labels:
        app: simpleapp-webapp
    spec:
      containers:
        - name: simpleapp-webapp
          image: docker.io/achalpadol/cloudethix_initcontainer_achal:v1
          ports:
            - containerPort: 80
```

```
-----
apiVersion: v1
kind: Service
metadata:
  name: simpleapp-service
spec:
  type: NodePort
  selector:
    app: simpleapp-webapp
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30009
```

```
-----
achal@LAPTOP-HPUOTJF1:kube$ kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
simpleapp-webapp-7c695998b5-4bmrw    1/1     Running   0           71s
simpleapp-webapp-7c695998b5-fx8ld    1/1     Running   0           71s
simpleapp-webapp-7c695998b5-l7wbt    1/1     Running   0           71s
```

```
-----
achal@LAPTOP-HPUOTJF1:kube$ kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP    PORT(S)
AGE
kubernetes                          ClusterIP       10.96.0.1        <none>          443/TCP
12h
simpleapp-service                    NodePort        10.106.134.224   <none>          80:30009/TCP
12s
```

```
-----
achal@LAPTOP-HPUOTJF1:kube$ minikube service simpleapp-service --url
http://127.0.0.1:38801
! Because you are using a Docker driver on linux, the terminal needs to be
open to run it.
```

```
-----
# updated deployment.yml file
achal@LAPTOP-HPUOTJF1:kube$ cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simpleapp-webapp
  namespace: cloudethix-achal

spec:
  replicas: 3
```

```
selector:
  matchLabels:
    app: simpleapp-webapp

template:
  metadata:
    labels:
      app: simpleapp-webapp

spec:

  initContainers:
  - name: install
    image: busybox:1.28
    command:
      - wget
      - "-O"
      - "/work-dir/index.html"
      - http://info.cern.ch
    volumeMounts:
      - name: workdir
        mountPath: "/work-dir"

  containers:
  - name: simpleapp-webapp
    image: docker.io/achalpadol/cloudethix_initcontainer_achal:v1
    ports:
      - containerPort: 80

    volumeMounts:
      - name: workdir
        mountPath: /usr/share/nginx/html

  volumes:
  - name: workdir
    emptyDir: {}
```

```
achal@LAPTOP-HPUOTJF1:kube$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
simpleapp-webapp-948549974-fxbf8	1/1	Running	0	103s
simpleapp-webapp-948549974-n5bbm	1/1	Running	0	103s
simpleapp-webapp-948549974-r76n8	1/1	Running	0	103s

```
achal@LAPTOP-HPUOTJF1:kube$ minikube service simpleapp-service --url -n
cloudethix-achal
http://127.0.0.1:42923
```

```
achal@LAPTOP-HPUOTJF1:~$ curl http://127.0.0.1:42923
```

```
<html><head></head><body><header>
<title>http://info.cern.ch</title>
</header>
```

```
<h1>http://info.cern.ch - home of the first website</h1>
```

```
<p>From here you can:</p>
```

```
<ul>
```

```
<li><a href="http://info.cern.ch/hypertext/WWW/TheProject.html">Browse the
first website</a></li>
```

```
<li><a href="http://line-
```

```
mode.cern.ch/www/hypertext/WWW/TheProject.html">Browse the first website using
```

```
the line-mode browser simulator</a></li>
<li><a href="http://home.web.cern.ch/topics/birth-web">Learn about the birth
of the web</a></li>
<li><a href="http://home.web.cern.ch/about">Learn about CERN, the physics
laboratory where the web was born</a></li>
</ul>
</body></html>
```

Que 6 →

- Create 1 Public Docker Hub registry named cloudethix_hpa_yourname.
- Clone below repository on your system.
https://github.com/vivekamin/kubernetes-hpa-example.git
- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , build a docker image from the docker file and add meaningful tags and push to the docker hub repository.
- Once the image is pushed, go to k8s directory and update deployment.yaml file with image name from your repo. And then create it.
- Open service.yml and change the type to nodePort and apply the same.
- Open the HPA.yaml file, notice it and then apply the same.
- Open the browser, and access the webpage.
- Now it's time to test the HPA working with the below command.
kubectl run -i --tty load-generator --rm --image=busybox --restart=Never -- /bin/sh -c "while sleep 0.01; do wget -q -O- http://NODE_PORT_SERVICE_NAME; done"
- Check the HPA from kubectl command and also check if the deployment is scaling up.
- Take the snap , prepare a well formatted doc and write your understanding.

```
achal@LAPTOP-# git clone https://github.com/vivekamin/kubernetes-hpa-
example.git
Cloning into 'kubernetes-hpa-example'...
remote: Enumerating objects: 26, done.
remote: Total 26 (delta 0), reused 0 (delta 0), pack-reused 26 (from 1)
Receiving objects: 100% (26/26), done.
Resolving deltas: 100% (9/9), done.
achal@LAPTOP-HPUOTJF1:~$ cd kubernetes-hpa-example
achal@LAPTOP-HPUOTJF1:kubernetes-hpa-example$ ls
Dockerfile README.md k8s package.json src
achal@LAPTOP-HPUOTJF1:kubernetes-hpa-example$
```

```
achal@LAPTOP-HPUOTJF1:cloudethix_hpa_achal$ cp -r ~/kubernetes-hpa-example/. .
achal@LAPTOP-HPUOTJF1:cloudethix_hpa_achal$
achal@LAPTOP-HPUOTJF1:cloudethix_hpa_achal$ ls
Dockerfile README.md k8s package.json src
```

```
achal@LAPTOP-HPUOTJF1:cloudethix_hpa_achal$ ls
```

```
Dockerfile README.md k8s package.json src
```

```
achal@LAPTOP-HPUOTJF1:cloudethix_hpa_achal$ sudo podman build -t
docker.io/achalpadol/cloudethix_hpa_achal:v1 .
STEP 1/6: FROM node:8.12.0-alpine
Resolved "node" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/node:8.12.0-alpine...
```

```
Getting image source signatures
Copying blob e35f88fcc259 done      |
Copying blob 4fe2ade4980c done     |
Copying blob eeb7d76f44e7 done     |
Copying config df48b68da0 done     |
Writing manifest to image destination
-----

achal@LAPTOP-HPU0TJF1:cloudethix_hpa_achal$ sudo podman images
REPOSITORY                                TAG          IMAGE ID
CREATED          SIZE
docker.io/achalpadol/cloudethix_hpa_achal  v1           f82e052dbcd4  45
seconds ago    69.2 MB
-----

achal@LAPTOP-HPU0TJF1:cloudethix_hpa_achal$ sudo podman push
docker.io/achalpadol/cloudethix_hpa_achal:v1
Getting image source signatures
Copying blob fd767799d266 done      |
Copying blob d2a386693d51 done     |
Copying blob 82fad7b4186f done     |
Copying blob eeb7d76f44e7 skipped: already exists
Copying blob e35f88fcc259 skipped: already exists
Copying blob 4fe2ade4980c skipped: already exists
Copying config f82e052dbc done     |
Writing manifest to image destination
-----

achal@LAPTOP-HPU0TJF1:k8s$ cat deployment.yml
achal@LAPTOP-HPU0TJF1:k8s$ cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: node-example
spec:
  replicas: 1

  selector:
    matchLabels:
      app: node-example

  template:
    metadata:
      labels:
        app: node-example
    spec:
      containers:
        - name: node-example
          image: docker.io/achalpadol/cloudethix_hpa_achal:v1
          imagePullPolicy: Always
          ports:
            - containerPort: 3000
          resources:
            limits:
              cpu: "0.5"
            requests:
              cpu: "0.25"
-----

achal@LAPTOP-HPU0TJF1:k8s$ kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
node-example-68f464997b-dczkj      1/1     Running   0           62s
-----

achal@LAPTOP-HPU0TJF1:k8s$ cat service.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: node-example
  labels:
    app: node-example
spec:
  selector:
    app: node-example
  ports:
    - port: 3000
      protocol: TCP
      nodePort: 30001
  type: NodePort
```

```
-----
achal@LAPTOP-HPUOTJF1:k8s$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP          13h
node-example        NodePort    10.103.35.120 <none>       3000:30001/TCP   11s
-----
```

```
# hpa before generating loads
achal@LAPTOP-HPUOTJF1:k8s$ kubectl get hpa
NAME                REFERENCE                TARGETS          MINPODS  MAXPODS
REPLICAS  AGE
node-example  Deployment/node-example  cpu: <unknown>/1%  1         4
0           2m57s
```

```
achal@LAPTOP-HPUOTJF1:k8s$ minikube service node-example --url
http://127.0.0.1:40001
! Because you are using a Docker driver on linux, the terminal needs to be
open to run it.
```

```
-----
achal@LAPTOP-HPUOTJF1:~$ curl http://127.0.0.1:40001
Hello World!!
```

```
# hpa after generating loads

^achal@LAPTOP-HPUOTJF1:k8s$ kubectl get hpa
NAME                REFERENCE                TARGETS          MINPODS  MAXPODS
REPLICAS  AGE
node-example  Deployment/node-example  cpu: 0%/50%     1         4         1
140m
```

```
-----
achal@LAPTOP-HPUOTJF1:k8s$ kubectl top pods
NAME                CPU(cores)  MEMORY(bytes)
node-example-68f464997b-dczkj  1m          25Mi
-----
```

- Que 7 →
- Create 1 Public Docker Hub registry named cloudethix_cronjob_yourname.
 - Initialize a local repository & copy below code (three files) to your local repository in any of your working branch.
 - Once code is copied, build the docker image from Dockerfile , add meaningful tags and then push the docker image to Docker hub registry.
 - Now update the pythoncronjob.yml file to change the image name that you have just pushed to docker hub registry.
 - Now create a cron job using pythoncronjob.yml file. Check with kubectl

command if the

cron job is created.

- Check the Job name which is created by cronjob from command line or lens.
- Then check the pod logs which are created by the job and capture the output.
- Prepare well formatted documents and write your understanding.

```
^Cachal@LAPTOP-HPUOTJF1:k8s$ cd
achal@LAPTOP-HPUOTJF1:~$ mkdir cloudethix_cronjob_achal
achal@LAPTOP-HPUOTJF1:~$ cd cloudethix_cronjob_achal/
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ ls
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ vim helloworld.py
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ vim Dockerfile
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ vim pythoncronjob.yml
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ ls
Dockerfile helloworld.py pythoncronjob.yml
```

Dockerfile

```
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ cat Dockerfile
FROM python:3.7-alpine
#add user group and ass user to that group
RUN addgroup -S appgroup && adduser -S appuser -G appgroup
#creates work dir
WORKDIR /app
#copy python script to the container folder app
COPY helloworld.py /app/helloworld.py
RUN chmod +x /app/helloworld.py
#user is appuser
USER appuser
ENTRYPOINT ["python", "/app/helloworld.py"]
```

helloworld.py

```
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ cat helloworld.py
#!/usr/local/bin/python3
import datetime
x = datetime.datetime.now()
print("Welcome to the Cloudethix World")
print("Today is")
print(x)
```

#cat pythoncronjob.yml

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: python-helloworld
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: python-helloworld
              image: docker.io/achalpadol/cloudethix_cronjob_achal:v4
              command: ["/app/helloworld.py"]
          restartPolicy: OnFailure
```

```
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ docker build -t
```

```
cloudethix_cronjob_achal:v4.
STEP 1/7: FROM python:3.7-alpine
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:3.7-alpine...
Getting image source signatures
Copying blob 96526aa774ef done      |
Copying blob 4819c95424fc done      |
Copying blob 9875af95546d done      |
Copying blob ea1518237b37 done      |
Copying blob 148762f75a1f done      |
-----

chal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ podman tag
cloudethix_cronjob_achal:v4 docker.io/achalpadol/cloudethix_cronjob_achal:v4
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ podman images
REPOSITORY                                     TAG          IMAGE
ID          CREATED          SIZE
localhost/cloudethix_cronjob_achal             v1
0e0635316d06  2 minutes ago  49.5 MB
docker.io/achalpadol/cloudethix_cronjob_achal   v1
0e0635316d06  2 minutes ago  49.5 MB
-----

achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ podman push
docker.io/achalpadol/cloudethix_cronjob_achal:v4
Getting image source signatures
Copying blob c43d3c0ede05 done      |
Copying blob 148762f75a1f skipped: already exists
Copying blob 4819c95424fc skipped: already exists
Copying blob 9875af95546d skipped: already exists
Copying blob ea1518237b37 skipped: already exists
Copying blob 96526aa774ef skipped: already exists
Copying blob b80bc22810df done      |
Copying blob 6aafb15c43ce done      |
Copying config 0e0635316d done      |
-----

achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ kubectl apply -f
pythoncronjob.yml
cronjob.batch/python-cronjob created
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ kubectl get cronjob
NAME          SCHEDULE          TIMEZONE    SUSPEND    ACTIVE    LAST SCHEDULE
AGE
python-cronjob */1 * * * *    <none>      False      0          <none>
31s

achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ kubectl get pods
NAME          READY    STATUS    RESTARTS    AGE
python-helloworld-29808324-f2jv8  0/1      Completed  0           102s
python-helloworld-29808325-wkpx5  0/1      Completed  0           42s
```